CASE STUDY 03

Audit Analytics and Automation: 25% Faster Data Preparation

n8n workflows and a Power BI model to pull and reconcile audit data for forensic and controls engagements.

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Client names and identifying details have been withheld. Prepared by Archish Chugh for portfolio purposes; this is not an EY publication and does not represent the views of any client.

CASE STUDY 03 · CROSS-SECTOR · ANALYTICS & AUTOMATION

Audit Analytics and Automation: 25% Faster Data Preparation

Set up n8n and Power BI workflows to pull and reconcile audit data 25% faster. This cut engagement preparation time and gave more time for actual control testing.

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USED ON	ROLE	TOOLS	RESULT
EY Risk & Compliance engagements	Designed and built the workflows	n8n · Power BI · Excel	25% faster data preparation

EXECUTIVE SUMMARY

SITUATION

Client data on forensic and controls engagements came in many formats: PDFs, ERP exports, registers and spreadsheets.

CHALLENGE

Collecting and matching this data by hand slowed every engagement and left less time for testing.

APPROACH

Built n8n workflows to collect and reconcile the data, added exception rules, and set up a Power BI model for review.

RESULT

Audit data pulled and reconciled 25% faster. Used on 2–3 engagements, with full-population testing where the data allows.

25%
faster data pull and reconciliation

2–3
engagements using the workflows so far

Full population
testing instead of samples, where the data allows

01

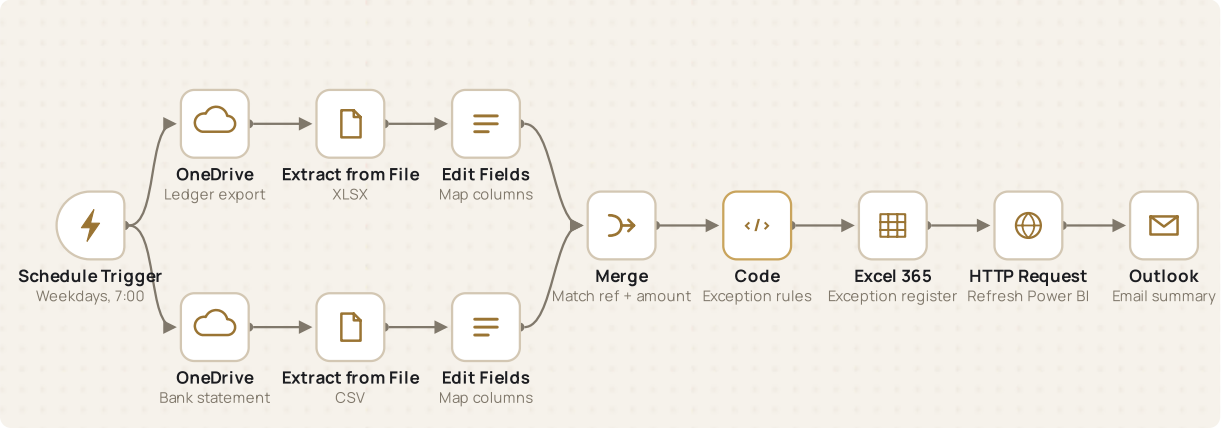
Background

On forensic and controls engagements, data arrives in many forms: bank statements as PDFs, ERP exports, agent sales files, collection registers and emailed spreadsheets. Analysts spent days collecting, reformatting and matching it by hand. This was slow, prone to errors and done differently on each engagement, and it reduced the time left for control testing.

02

n8n Workflow

Exhibit 1: n8n Workflow Canvas Representative design. Hover over a node for detail.



STEP	N8N NODE	WHAT IT DOES
1. Start	Schedule Trigger	Runs the workflow every weekday morning.
2. Collect	OneDrive, Extract from File	Downloads the ledger export (XLSX) and bank statement (CSV) from the engagement folder and reads the rows.
3. Standardise	Edit Fields	Maps both sources to the same columns: date, reference and amount.
4. Match	Merge	Matches ledger and bank lines on reference and amount.
5. Test	Code	Applies the exception rules (unmatched items, amount mismatches, round amounts, weekend postings) and passes on only the exceptions.
6. Report	Excel 365, HTTP Request, Outlook	Adds exceptions to the review register, refreshes the Power BI dataset through its REST API, and emails the team a summary.

The same workflow pattern is used for other reconciliations, such as sales records against agent records, by changing the source files and the matching keys. Unmatched items then become the starting point for control testing rather than something found at the end.

03

Working Demo

This is a smaller version of the matching and exception rules used in the workflow. It uses sample data, not client records. Turn rules on or off to see how the exception list changes.

Exhibit 2: Collections, Books Against Bank (Sample)

Sample data for demonstration

9 items tested **3** matched **6** exceptions

DATE	REFERENCE	BOOKS ₹	BANK ₹	RESULT
Mon, 03 Mar	Agent A collection	48,250	48,250	Matched
Tue, 04 Mar	Agent B collection	1,20,000	1,20,000	Round amount
Wed, 05 Mar	Counter sales	36,780	35,780	Differs by ₹1,000
Sat, 08 Mar	Cash deposit CD-114	25,500	25,500	Weekend posting
Mon, 10 Mar	Agent C collection	62,410	—	Not in bank
Tue, 11 Mar	Agent A collection	51,090	51,090	Matched
Wed, 12 Mar	Agent E collection	50,000	50,000	Round amount
Thu, 13 Mar	Unidentified receipt	—	18,600	Not in books
Fri, 14 Mar	Agent D collection	44,375	44,375	Matched

04

Power BI Data Model

Exhibit 3: Star Schema Behind the Dashboards

Representative design

Dim_Entity
Legal entities, directors, shared addresses

Fact_Transactions
Every ledger and bank line, with a match status and exception flags

Dim_Counterparty
Vendors, agents and customers, with related-party tags

Dim_Date
Period, fiscal year, weekend and holiday flags

Fact_Exceptions
Rule hits, reviewer, disposition and evidence reference

Dim_Control
Control ID, owner, COSO component, test result

The dashboards had three views. An engagement overview showed reconciliation coverage and open exceptions. An exception register let reviewers filter by rule, entity and value and record what they decided. A controls view

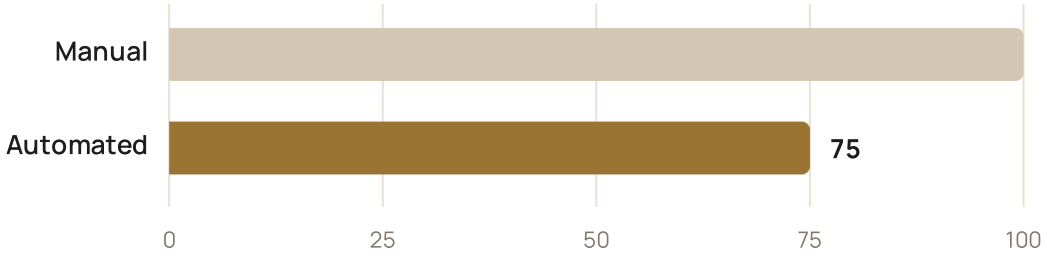
linked each exception to the control that should have prevented it, so the audit report could go straight from the data to the root cause.

05

Results

Exhibit 4: Time to Pull and Reconcile Engagement Data

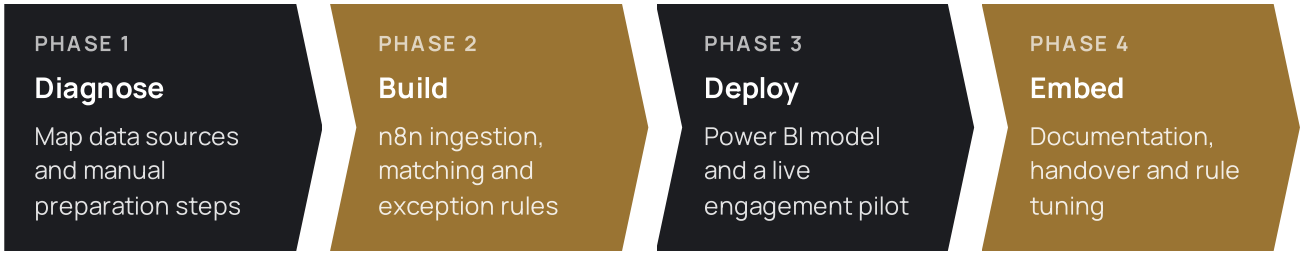
Index: manual process = 100; lower is faster



- **Productivity:** audit data was pulled and reconciled 25% faster, and the time saved went into control testing.
- **Assurance:** reconciliations cover full populations, so every transaction is tested instead of a sample and exceptions cannot fall between samples.
- **Consistency:** the same rules run on every engagement, and each exception links to its evidence reference for review.

06

Rollout



1 **Piloted on a live engagement**
Built the first workflow on a live engagement, so the time saved was measured on real work.

2 **Walked managers through the exceptions**
Walked managers through the exception register and how each flag ties back to a control finding.

3

Documented the workflows

Documented each workflow's inputs, rules and outputs so other analysts can reuse it independently.

4

Improved the rules from feedback

Added and adjusted rules based on reviewers' decisions on each exception.

KEY TAKEAWAYS

1. Automating data preparation gives more time for control testing.
2. Rules run on the full population catch exceptions that samples can miss.
3. A live pilot and clear documentation helped the team start using the workflows.

UAE

Relevance to UAE Roles

Internal audit teams in UAE banks, airlines and government entities are adding data analytics and continuous auditing to their work, in line with IIA Global Internal Audit Standards. This experience can be applied with whichever automation or BI tools a team already uses.